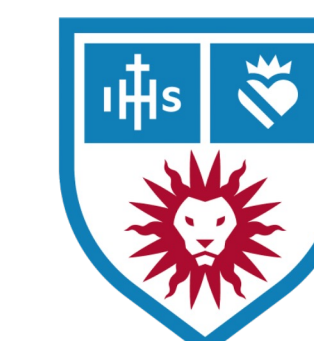
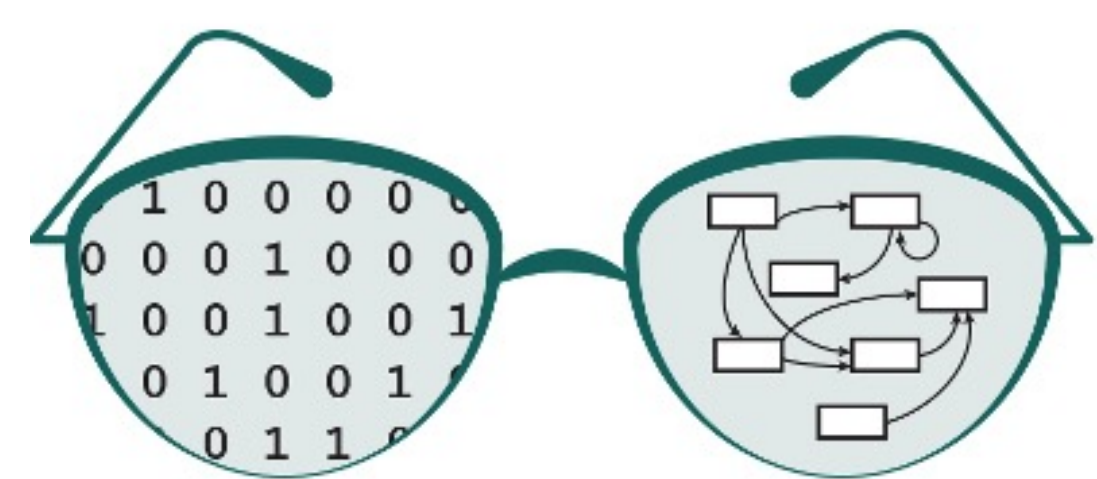


Database, Display, and DevOps Improvements for GRNsight 7.2, a Web Application for Visualizing Gene Regulatory and Protein-Protein Interaction Network Models



Loyola Marymount University



Ngoc K. Tran¹, Cecilia J. Zaragoza¹, A'Kaia L. Phelps¹, Amelie T. Dinh¹, Milka Y. Zekarias¹, Kam D. Dahlquist², John David N. Dionisio¹
Departments of ¹Computer Science and ²Biology, Loyola Marymount University, 1 LMU Drive, Los Angeles, CA 90045 USA

<http://dondi.github.io/GRNsight/>

GRNsight, automatically lays out a gene regulatory network (GRN) graph, coloring the edges based on regulatory weights, and the nodes based on experimental or simulated expression data

1. Network

- Demo files are provided.
- Can import and export Excel, SIF, or GraphML files.
- Can construct a network from SGD regulation or PPI data stored in a backend database.

2. Layout

- Users can select a force graph layout or grid layout.
- Force graph parameter sliders:
 - Link distance determines the minimum distance between nodes.
 - Nodes have a charge which repels or attracts other nodes.

3. Node

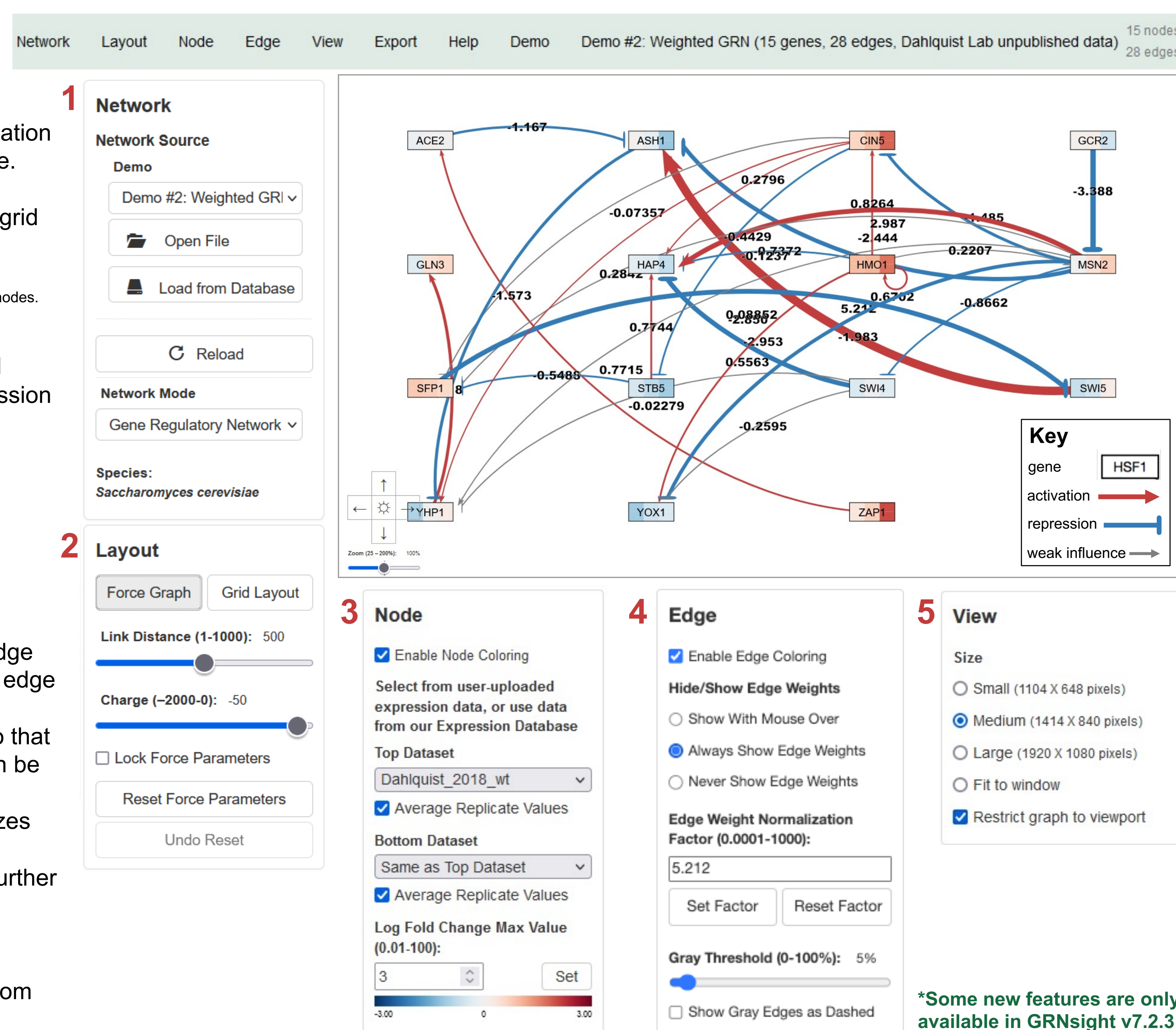
- Coloring can be based on user-uploaded expression data or from published expression datasets stored in a backend database.
- Datasets available:
 - Apweiler et al. 2012, GSE33098
 - Barreto et al. 2012, GSE24712
 - Dahlquist et al. 2018, GSE83656
 - Kitagawa et al. 2002, GSE9336
 - Thorsen et al. 2007, GSE6068

4. Edge

- Edge coloring can be toggled on and off.
- Buttons enable the user to always see edge weights, never see edge weights, or see edge weights upon mouseover.
- Allows user to set normalization factor so that edge thicknesses for different graphs can be rendered on the same scale.
- Changing the gray threshold deemphasizes weaker edges in the visualization.
- Presenting gray edges as dashed lines further deemphasizes weak edges.

5. View

- Multiple viewport sizes available.
- Graph bounding box can be separated from viewport.
- Zooming and scrolling available.

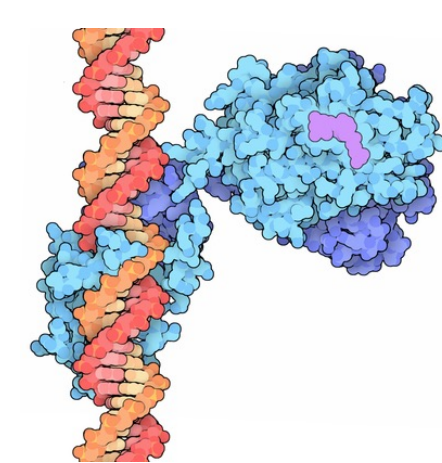


*Some new features are only available in GRNsight v7.2.3 beta.

GRNsight can now be used to visualize protein-protein physical interaction (PPI) networks

- A gene regulatory network consists of genes, transcription factors, and the regulatory connections between them which govern the level of expression of mRNA and protein from genes.

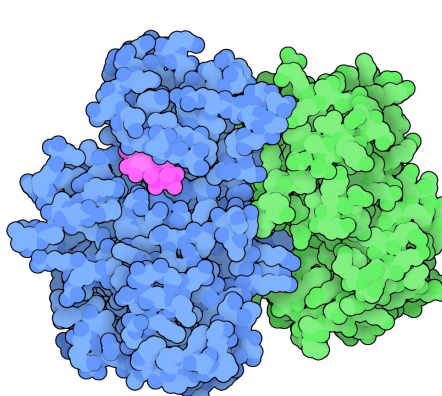
- Nodes represent both the regulatory transcription factor protein and the gene that encodes it.
- An arrow represents a regulatory transcription factor either activating (arrowhead, red color) or repressing (blunt end, blue color) the gene for another regulatory transcription factor.



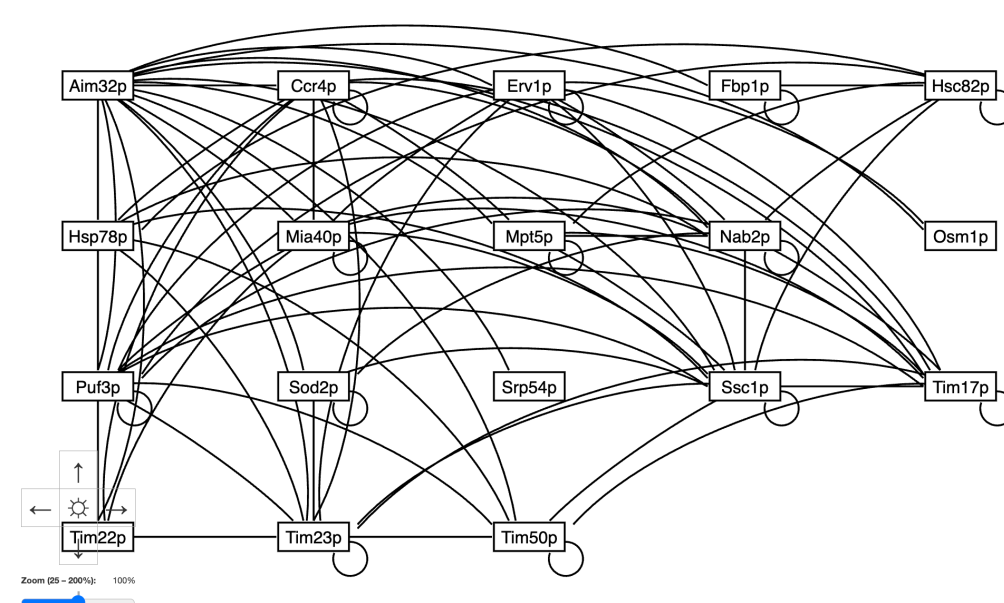
Regulatory Transcription Factor Bound to DNA
Image from: <https://pdb101.rcsb.org/motm/155>

- A PPI is when two proteins bind to each other.

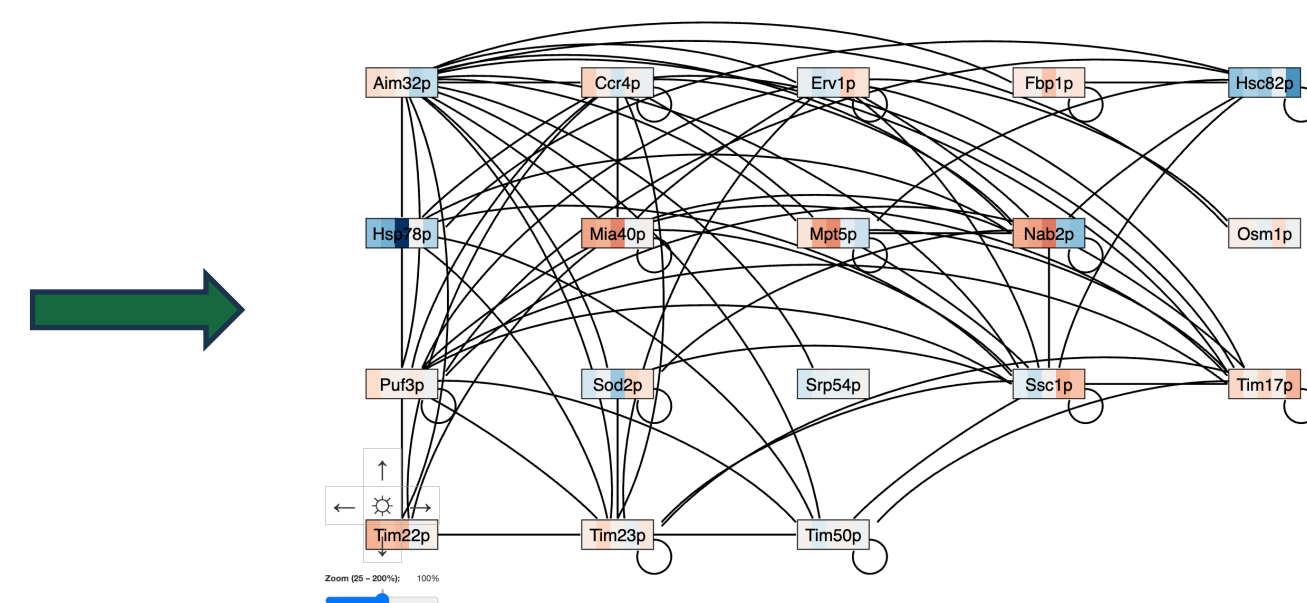
- Each node represents a protein, and each edge represents their binding interaction.
- Unlike GRNs, PPI graphs have undirected edges of uniform thickness and black color and previously did not display expression data on the nodes.



Protein-Protein Interaction
Image from: <https://pdb101.rcsb.org/motm/236>



Demo #5: PPI with Node Coloring Disabled



Demo #5: PPI with Node Coloring Enabled

Node Coloring Support for PPI Networks

1. Why PPI Node Coloring?

- Previously, node coloring was only enabled for GRNs because node coloring is a visualization of the level of mRNA expression using data provided by the user or from GRNsight's expression database.

2. PPI Node Coloring Implementation

- Now, when a PPI demo or network without expression data in the workbook is loaded, node coloring defaults to disabled.
- When node coloring is enabled for a PPI, a warning message displays that informs users mRNA-level expression data is being displayed on a PPI network.
- The query on the expression database for gene names required a "p" at the end of the gene name to account for the proper protein nomenclature.

Background & Motivation Behind Latest Updates

- Our prior addition of PPI support has not yet permeated the app fully, and we have an on-going task of integrating PPIs into more of the app's functionality.
- YeastMine, the original provider of our network and expression data, was shut down, necessitating a search for a new data source, AllianceMine.
- Travis CI, our original provider for continuous integration/development (CI/CD) services, ended free support.
- GRNsight's increasing functionality resulted in a range of UI loose ends and bugs that need to be addressed in order to support a wider audience.

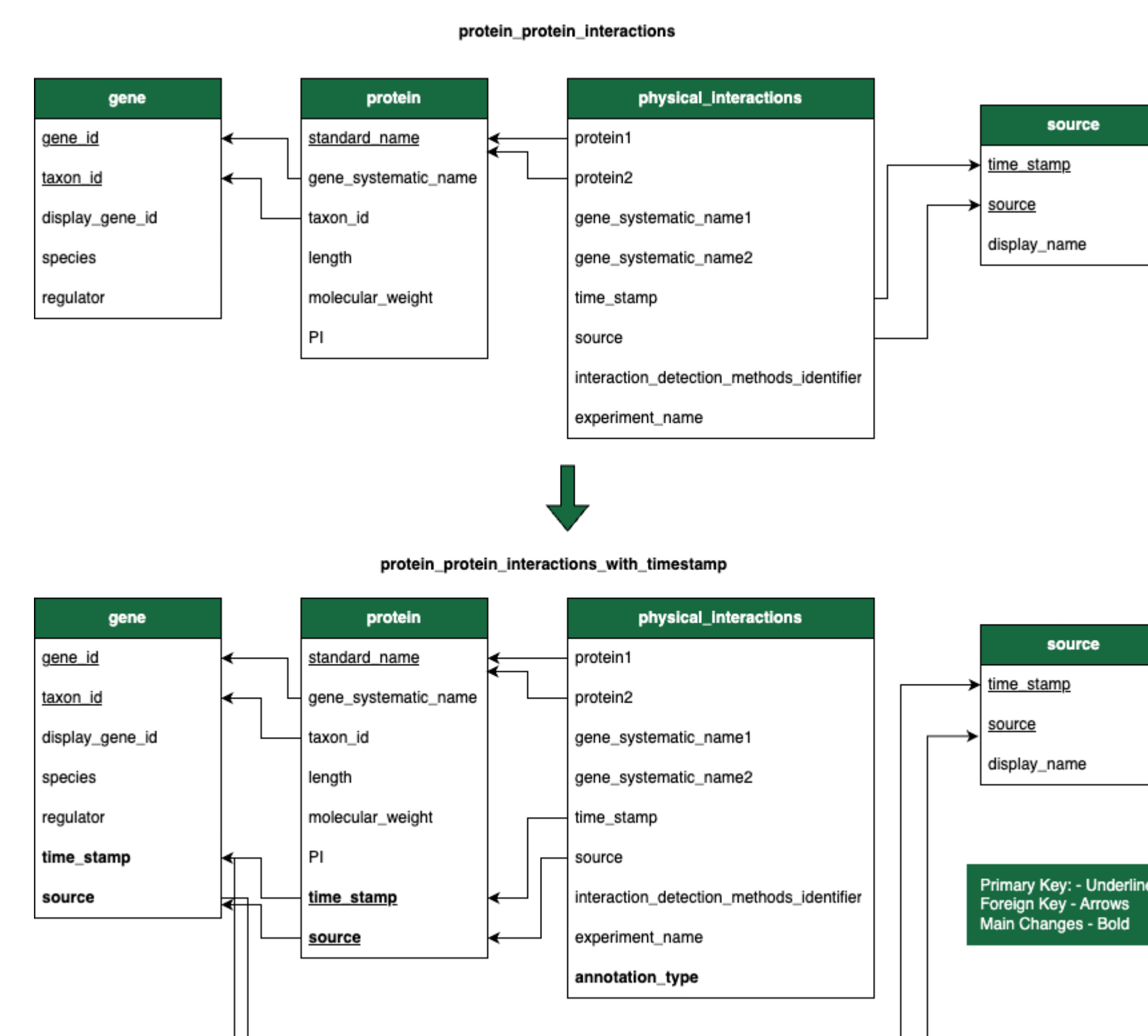
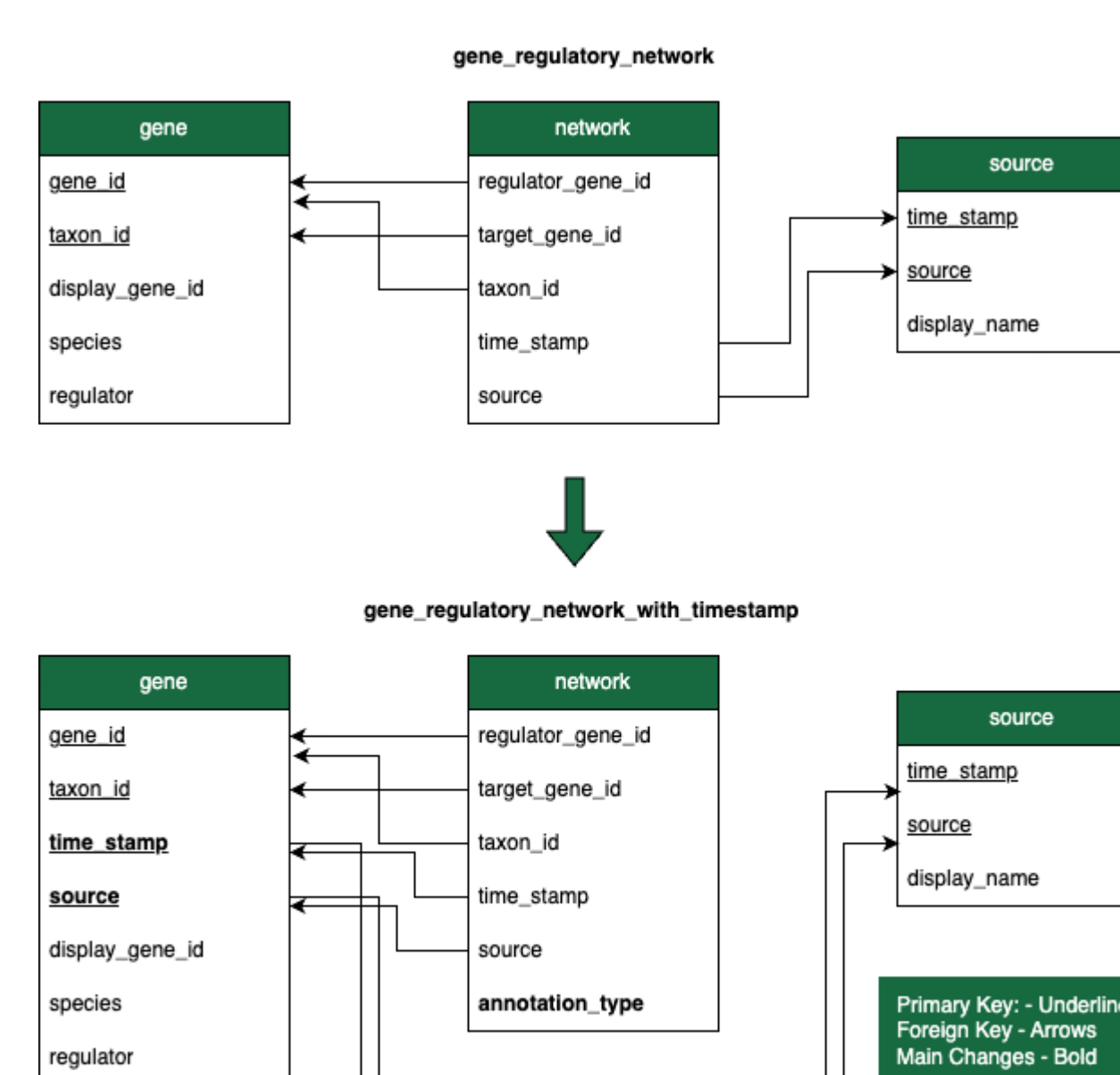
New GRNsight database source and schema

1. New Database Source: AllianceMine

- GRNsight has transitioned to AllianceMine as our new database source, replacing YeastMine, which is currently unavailable.

2. New Schemas

- GRNsight has updated its database schema to include:
 - Timestamps for gene and protein tables** – Each gene and protein entry now includes a timestamp, allowing us to distinguish data across different yearly database updates. If a gene does not exist within a specific timestamp, it cannot be queried, ensuring clear separation of data from different updates.
 - Annotation type** – Each regulatory relationship is either manually curated from the literature or from high-throughput experiments. Eventually, there will be the ability to select whether to look at manually curated, high-throughput, or both types of interactions on GRNs and PPIs that are loaded from the database



Migrating the CI/CD Pipeline

1. Transitioning GRNsight's Pipeline

- Transitioned GRNsight's CI/CD pipeline from Travis CI to GitHub Actions and integrated automated workflows for builds, unit tests, and linting to streamline development and enhance maintainability.

2. Impact of These Changes:

- Developers do not have to manually run tests or check for errors, GRNsight is easier to maintain and update since potential issues are detected early, and transitioning to a modern pipeline will ensure compatibility with future JavaScript versions and improve performance.

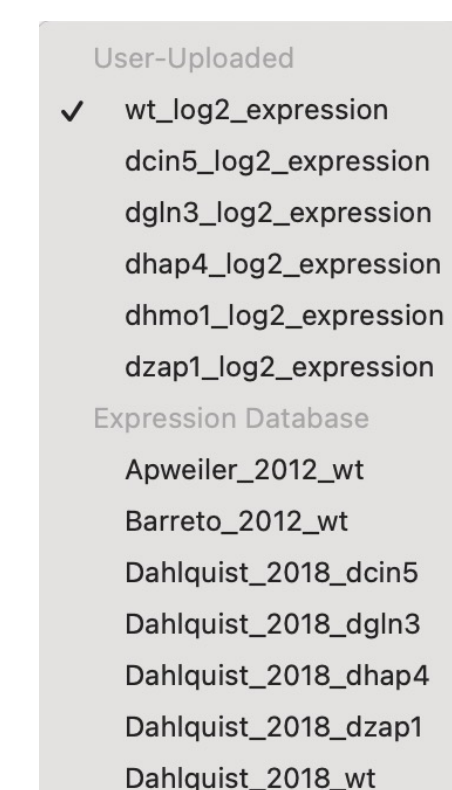
User Interface Dropdown Bug Resolutions

1. Clear Differentiation Between User-Uploaded Data and Expression Database

- The data options in the dropdown menu were separated into two different opt-groups: User-Uploaded data and Expression Database data.

2. Updating Demo Dropdown

- The demo dropdown was updated to align with proper code structure and protocol, improving the resetting functionality when uploading custom workbooks.
- A demo dropdown grnstate value was created to refer to the workbook state, connecting the demo dropdown value and the front facing user interface display.



Graph View Controls

1. Enabling Controls of D-pad, Zoom Slider, and Cursor Grab When Restrict Graph to Viewport is Selected

- The D-pad, Zoom Slider, and Cursor Grab controls can be utilized while ensuring the nodes remain clamped inside the bounding box of the viewport, outlined in gray, which enhances interaction with the graph.
- Previously, nodes would be restricted to the chosen viewport size of the graph, which limited interaction with the graph.

2. Fit Graph to Window

- The size of the bounding box that contains the nodes can dynamically adjust to fit the bottom and right edges of a browser window.
- Previously, when Fit to Window was selected, the graph would shrink to the size of one node because the asynchronous communication between GRNsight and the iFrame it is contained within causes a delay, so the graph was being loaded with undefined dimensions.

Future Directions

- Prioritize bug resolution to enhance platform stability and reliability.
- Improve documentation for clearer and more comprehensive user guidance.
- Publish a research paper detailing GRNsight's advancements and methodologies since the prior publication.
- Migrate to React to modernize the platform and improve performance.

Availability

- GRNsight is free and open to all users at <http://dondi.github.io/GRNsight/>, and there is no login requirement.
- GRNsight code is available under the open source BSD license at our GitHub repository at <https://github.com/dondi/GRNsight>.
- Some new GRNsight features are only available in GRNsight v7.2.3 beta.

Acknowledgments

We would like to thank Nikki C. Chun and the previous developers on the GRNsight project. A'Kaia Phelps and Cecilia J. Zaragoza would like to thank the HHMI STEM Research Learning Community for funding.

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